



OKW Data analysis

This data analysis was made using the next tools:

Open Refine, Python with Pandas, and Datasette. The maps are intended to show the distribution of the data collected, but due to the format it has some limitations.

The next table of data shows entries by country, machines, facilities, and most common machine.

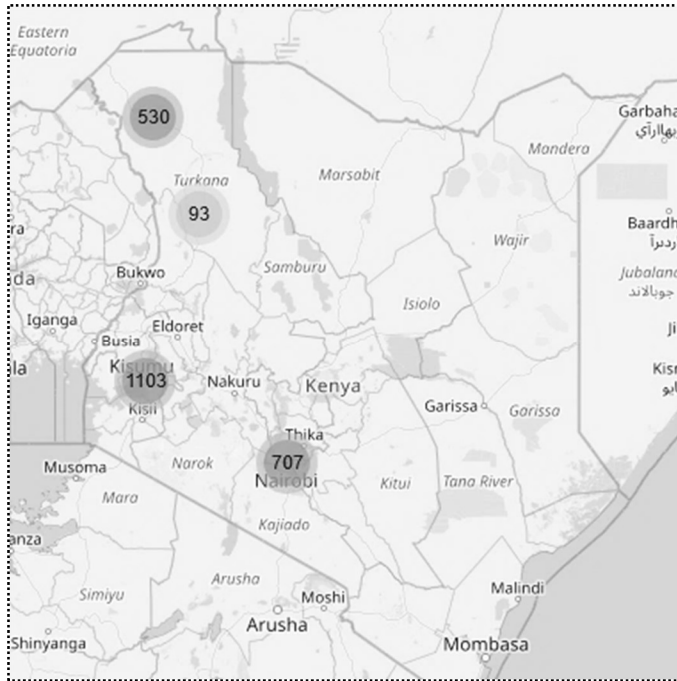
Country	Machines	Facilities	Most common machine	Qty	%
Kenya	3232	2569	Sewing machine	1334	41
Uganda	2832	2626	Welding Rod	752	40
Bangladesh	719	442	Lathe	335	46
Somalia	118	45	3D printer	12	10
Somaliland	61	24	Mixer	19	80
Congo DRC	39	39	Sewing Machine	19	31
Iraq	24	17	Laser cutter	7	30
Burundi	3	3	Sewing Machine	3	
Ethiopia	1	1	Sewing Machine	1	

In the next two pages we show data from Kenya and Uganda. In Kenya the most common tool in our database is the Sewing machine and in Uganda is the Welding Rod. It may be an indication of the people that was more open to answer the survey but also about the more important economic activities in the region.



KENYA

7 Counties
31 Sub Counties

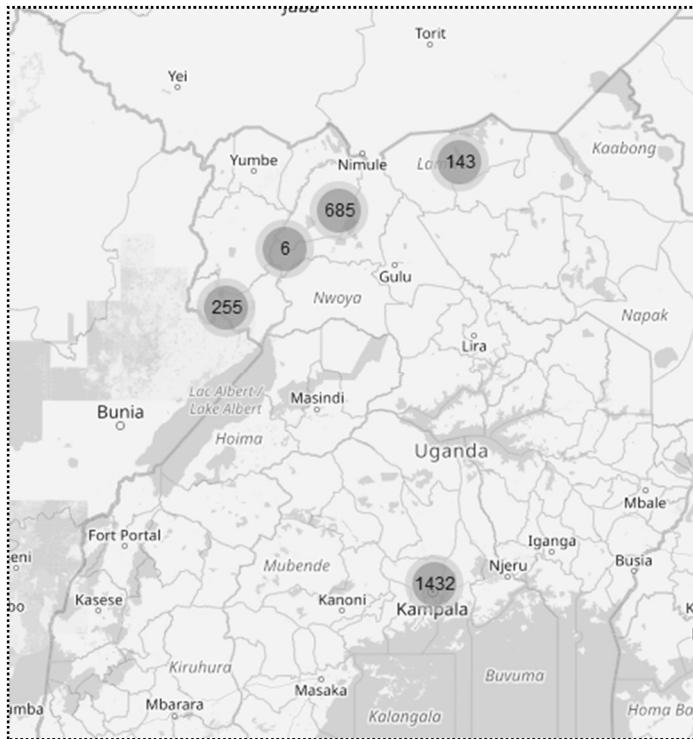


Data entries per cities	
Kisumu	1240
Nairobi City	775
Turkana	642
Kwale	5
Kilifi	2
Mombasa	1
Taita Taveta	1
3232	
2667	

By counties	
Kisumu Central	522
Turkana West	474
Kisumu East	248
Embakasi	176
Kisumu West	167
Dagoretti	132
Mathare	132
Kibra	106
Nyando	106
Turkana South	95
Starehe	94
Nyakach	86
Kamukunji	73
Turkana Central	70
Muhoroni	49
Seme	43
Embu_West	33
Kasarani	15
Makadara	13
Homa_Bay	10
Kisii_Central	8
Baringo_Central	2
Embu_North	2
Koibatek	2
Turkana_East	2
Baringo_North	1
East_Pokot	1
Kakamega_Central	1
Kibwezi	1
Marigat	1
Turkana_North	1

Machine types				
Sewing Machine	1334	56%	Press Brake	5
Grinding Machine	430	18%	Vinyl Cutter	5
Welding Rod	302	12%	Injection Molding	3
Mixer	94	4%	3D Printer FDM	2
Embroidery Machine	91	4%	3D Printer SLA	2
CNC Router	42	6%	3D Scanner	2
Laser Cutter	24		CNC Punching	2
CNC Milling	12		Mill	2
Lathe	10		3D Printer SLS	1
Heating Press	8		Plasma Cutter	1

UGANDA



Data entries per cities	
Wakiso	1231
Adjumani	406
Mukono	403
Amuru	315
Nebbi	176
Lamwo	150
Zombo	122
Kampala	3
Arua	2
Masajja	2
Lira	1
Madi Okollo	1

Places	
2626	

Machines	
2832	

Materials used	
331	Steel
151	Stainless Steel

Machine types				
Welding Rod	752	40%	Injection Molding	2
Sewing Machine	716	38%	3d Printer Sla	1
Grinding Machine	334	18%	CNC Milling	1
Laser Cutter	22	4%	CNC Punching	1
Mill	15		CNC Router	1
Plasma Cutter	8		Heating Press	1
Vinyl Cutter	8		Lathe	1
Embroidery Machine	2		Mixer	1

Coberture	
414	Parishs
12	Districts

COVID-19 context:

We can see a direct relationship between the number of sewing machines and the increase in manufacturing of facemasks as an effect of the pandemic.

This table shows the main products of every facility using sewing machines:

Face mask	601	43%
Clothing	254	18%
Face mask, clothing	139	10%
Face mask, clinical coats	122	9%
Face mask, uniform, clothing	80	6%
Face mask, clothing, Porches, Aprons, Bags	156	11%
Tailoring	42	3%
Textiles	14	1%

Approximately 78% of all the owners of Sewing machines manufactured Facemasks. Using data from all countries on the database.

To find a live version of this document use the next URL:

https://github.com/iop-alliance/okw_data_wrangling/blob/main/OKW_data_analysis_05_2022.pdf